

Topic 12

Patterns in the Chemical World

Chemistry in Context

The latest rechargeable cell — Vanadium Redox Battery When we think of batteries, we tend to think of small things, but what if you want to store lots of power? Then you need a huge battery, and that is why the vanadium redox battery is developed. Indeed, the vanadium redox battery is designed to have a large-scale electricity storage capacity. Vanadium is a transition metal. It is able to exist in solution in different oxidation states. A vanadium redox battery consists of an assembly of cells in which two electrolytes are separated by a membrane. The electrolyte in the cathode compartments contains $VO_2^+(aq)$ and $VO^{2+}(aq)$ ions while the electrolyte in the anode compartments contains $V^{3+}(aq)$ and $V^{2+}(aq)$ ions. The battery can offer almost unlimited capacity simply by using larger and larger storage tanks for the electrolytes. Moreover, it can be left completely discharged for long periods with no negative effects and can be recharged simply by replacing the electrolyte if no power source is available to charge it.

Do you recall?

1 When a vanadium redox battery discharges, $VO_2^+(aq)$ ions in the cathode compartments are converted to $VO^{2+}(aq)$ ions. Write an ionic half-equation for the reaction that occurs.

$VO_2^+(aq) + 2H^+(aq) + e^- \longrightarrow VO^{2+}(aq) + H_2O(l)$

2 In terms of oxidation state, explain whether the $VO_2^+(aq)$ ions in the cathode compartments are oxidized or reduced.

$VO_2^+(aq)$ ions are reduced. It is because the oxidation number of vanadium decreases from +5 to +4.

3 What is the major problem in using a lead-acid battery?

Lead is a highly poisonous metal.

In this topic, you are going to study the periodic trends for elements in the periodic table and transition metals.

Unit

41 Periodic trends in elements and their compounds

Pre-study

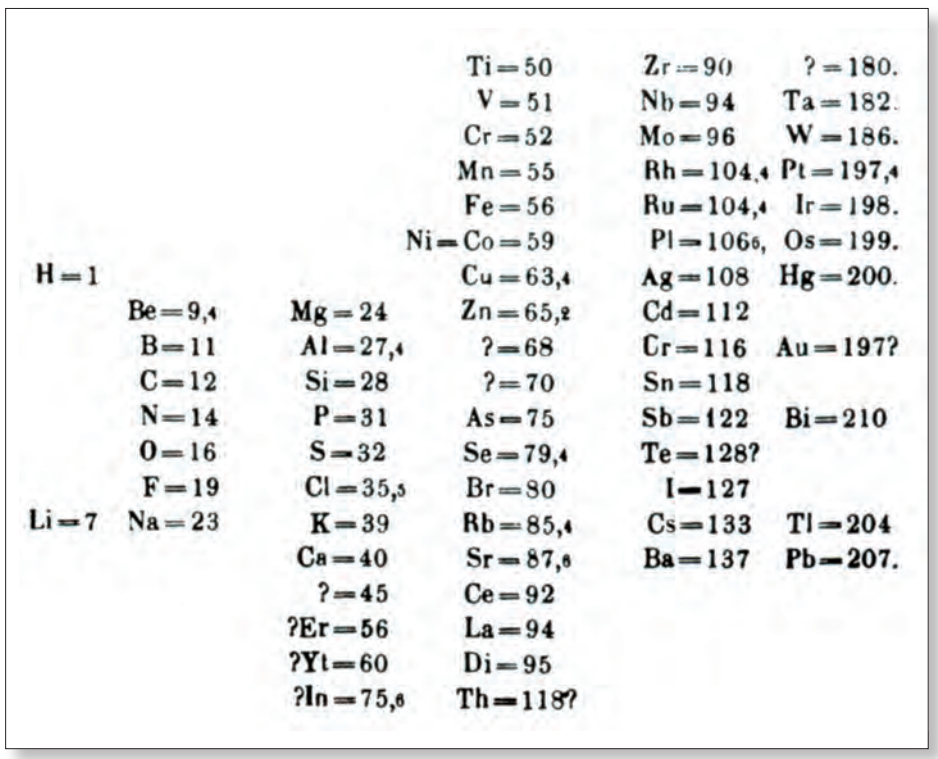
A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

- Periodic variations in structures and bonding of elements across Periods 2 and 3
- Variations in melting point and electrical conductivity of elements across a period
- Bonding, composition and acid-base nature of oxides of elements across Period 3

41.1 Development of the periodic table

The first periodic table was published by the Russian chemist Mendeléev in 1869 (Fig. 41.1). He arranged the elements in order of relative atomic mass. We now know that the elements in the periodic table are arranged in order of atomic number. Fortunately, these two arrangements — in order of atomic number and of relative atomic mass — are almost identical.



			Ti = 50	Zr = 90	? = 180.
			V = 51	Nb = 94	Ta = 182.
			Cr = 52	Mo = 96	W = 186.
			Mn = 55	Rh = 104,4	Pt = 197,4
			Fe = 56	Ru = 104,4	Ir = 198.
			Ni = Co = 59	Pl = 106,6	Os = 199.
H = 1			Cu = 63,4	Ag = 108	Hg = 200.
	Be = 9,4	Mg = 24	Zn = 65,2	Cd = 112	
	B = 11	Al = 27,4	? = 68	Ur = 116	Au = 197?
	C = 12	Si = 28	? = 70	Sn = 118	
	N = 14	P = 31	As = 75	Sb = 122	Bi = 210
	O = 16	S = 32	Se = 79,4	Te = 128?	
	F = 19	Cl = 35,5	Br = 80	I = 127	
Li = 7	Na = 23	K = 39	Rb = 85,4	Cs = 133	Tl = 204
		Ca = 40	Sr = 87,6	Ba = 137	Pb = 207.
		? = 45	Ce = 92		
		?Er = 56	La = 94		
		?Yt = 60	Di = 95		
		?In = 75,6	Th = 118?		

Fig. 41.1 Mendeléev's periodic table

The vertical columns of Mendeléev's periodic table correspond to the periods of elements in the modern periodic table and the horizontal rows are the groups of elements.

Worksheet Assessment Form
A worksheet and an assessment form on this activity are available for download from *Jing Kung Chemistry Website*.

Find & Share

The growing periodic table

The periodic table has undergone significant changes since Mendeléev published his first version in 1869. Henry Moseley (Fig. 41.2), who determined the atomic numbers of elements, and Glenn Seaborg (Fig. 41.3), who synthesized *transuranium elements* (the elements after uranium in the periodic table), are two major contributors to developments of the periodic table.



Fig. 41.2 Henry Moseley

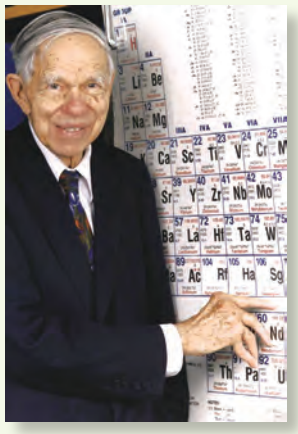


Fig. 41.3 Glenn Seaborg

Task

You are going to search for information on the contributions of Moseley and Seaborg to the development of the periodic table. Your teacher will divide the class into groups. Each group should prepare a poster to present the findings.

Hints for the search

1 What technique did Moseley use to determine the atomic numbers of elements?

2 How was the periodic law proposed by Mendeléev modified by Moseley?

3 How many transuranium elements were synthesized by Seaborg?

4 How was the periodic table modified by the work of Seaborg?

Refer to page T9.

Reference websites

1 You may start your search from the following websites:

- Infoplease.com — a brief history of the periodic table
  http://www.infoplease.com/ipa/A0905215.html
- Royal Society of Chemistry — information about the development of the periodic table
  http://www.rsc.org/periodic-table/history
- Princeton University — information about Glenn Seaborg
  http://www.princeton.edu/~achaney/tmve/wiki100k/docs/Glenn_T._Seaborg.html

2 Extend your search using search engines and appropriate keywords.

transuranium element 鈾後元素

41.2 Categories of elements in the modern periodic table

Elements in the modern periodic table (Fig. 41.4) are often divided into four categories:

1 **Main group elements**

2 **Transition metals**

3 **Lanthanides**

4 **Actinides**

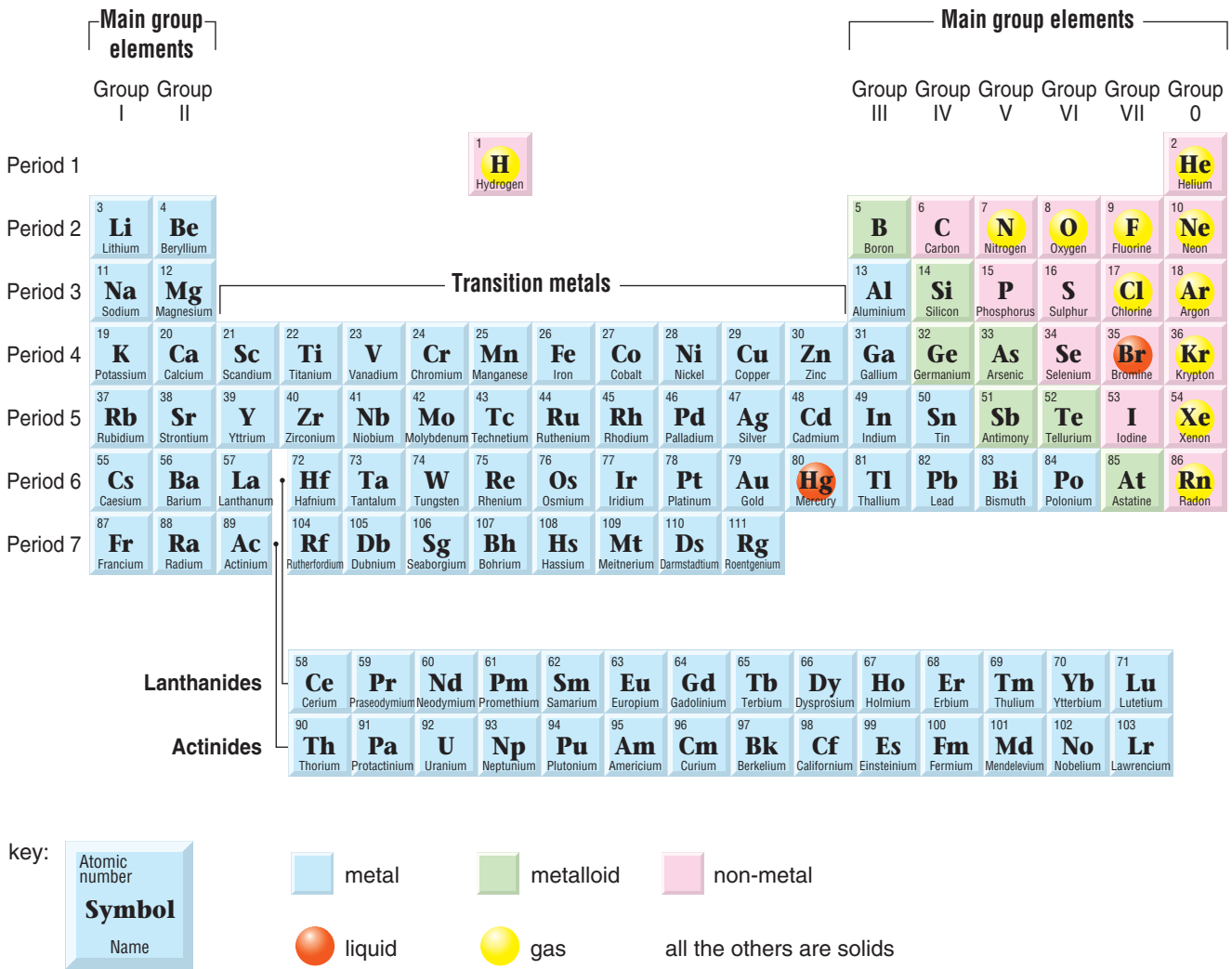


Fig. 41.4 The modern periodic table

main group element 主族元素　　transition metal 過渡金屬　　lanthanide 鑭系元素　　actinide 錒系元素

Metalloids refer to those elements that exhibit some of the properties of metals and non-metals.

The properties of elements vary in a fairly regular way as we move across a period from left to right. For example, every time we go across a period, we go from metals on the left through metalloids to non-metals on the right (Fig. 41.5). This recurrence of the same pattern is called **periodicity**.

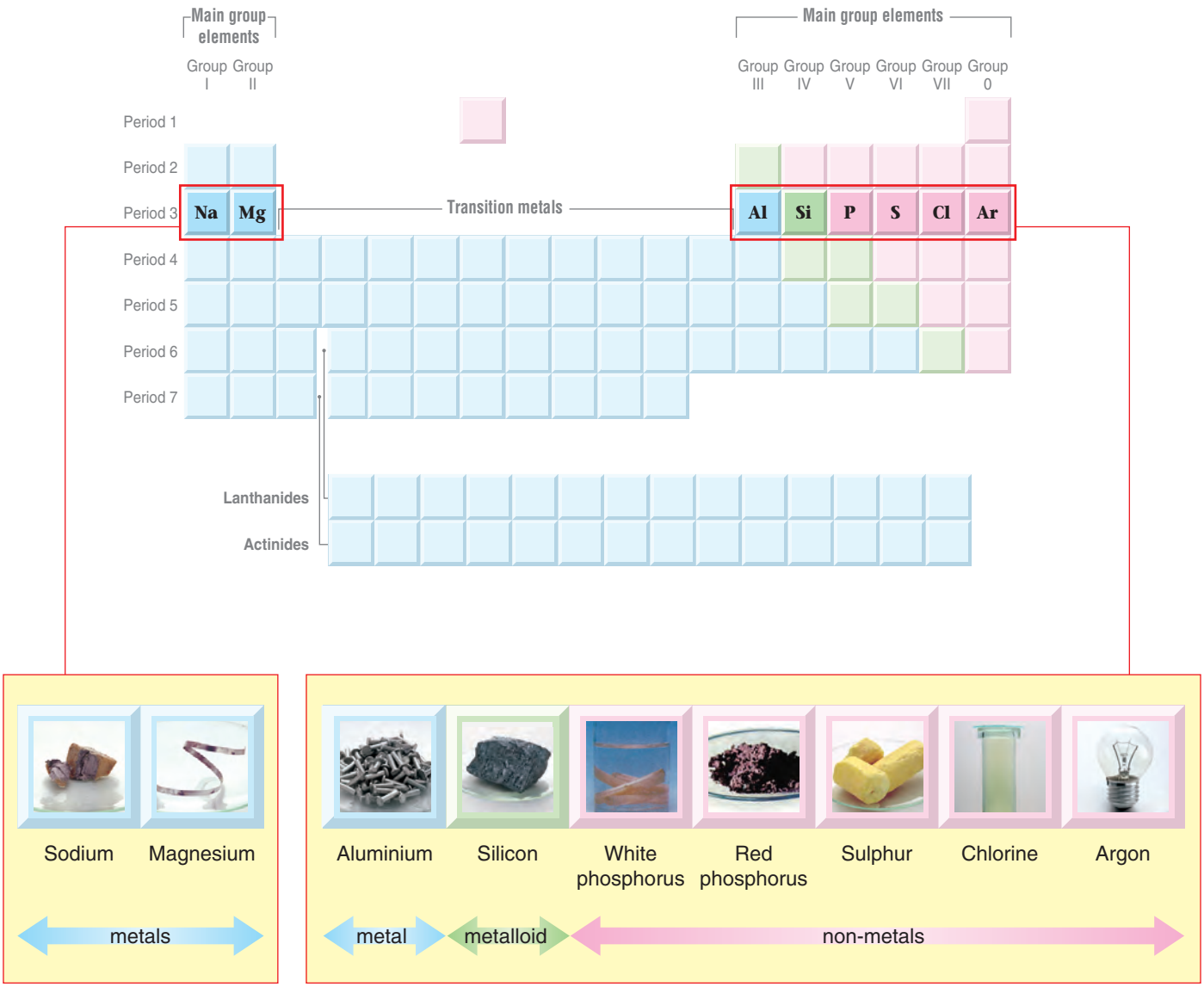


Fig. 41.5 Samples of the elements in Period 3

periodicity 週期律

41.3 Periodic variations in the type of structure and bonding of elements across periods 2 and 3

Neon and argon exist as monoatomic molecules.

Table 41.1 shows the variation in the type of structure and bonding of elements across periods 2 and 3. Structures of the elements vary from metallic in the metals, through giant covalent in the metalloids to simple molecular in the non-metals.

Table 41.1

Variation in the type of structure and bonding of elements across periods 2 and 3

Group	I	II	III	IV	V	VI	VII	0
Element in Period 2	Li	Be	B	C (diamond)	N	O	F	Ne
Type of element	metals		metalloid	non-metals				
Type of structure	**giant metallic**		**giant covalent**		**simple molecular**			
Types of bonding and intermolecular forces	metallic		covalent		covalent bond holding atoms together within the molecule; **van der Waals' forces** between molecules			van der Waals' forces between monoatomic molecules
Element in Period 3	Na	Mg	Al	Si	P	S	Cl	Ar
Type of element	metals			metalloid	non-metals			
Type of structure	giant metallic			giant covalent	simple molecular			
Types of bonding and intermolecular forces	metallic			covalent	covalent bond holding atoms together within the molecule; van der Waals' forces between molecules			van der Waals' forces between monoatomic molecules

giant metallic structure 巨型金屬結構　giant covalent structure 巨型共價結構　simple molecular structure 簡單分子結構
van der Waals' forces 范德華力

Giant metallic structure

ASL 2009 Paper 1 Q9
AL 2010 Paper 1 Q9

We can describe the metallic bonding in a metal as positive metallic ions arranged in a lattice held together by a 'sea' of **delocalized electrons** (Fig. 41.6). The electrostatic attractions between the positive metallic ions and the 'sea' of electrons bind the entire structure together as a single unit.

The metallic bonding spreads throughout, so the metal has a giant structure.

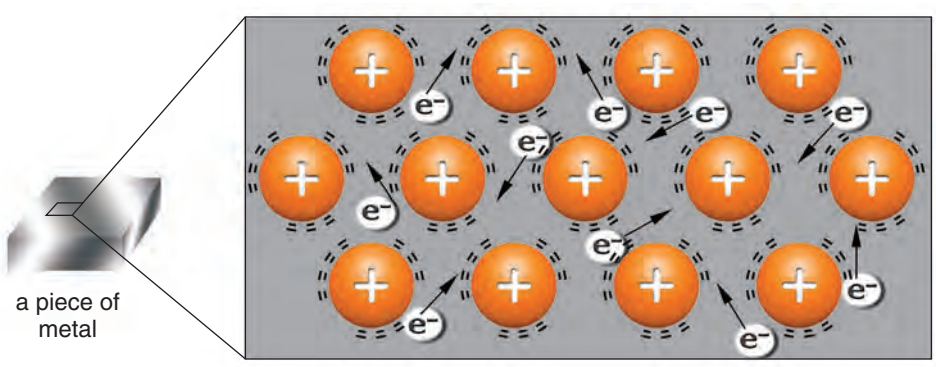


Fig. 41.6 A piece of metal consisting of positive metallic ions surrounded by a 'sea' of delocalized electrons

N1 Germanium has a giant covalent structure as well.

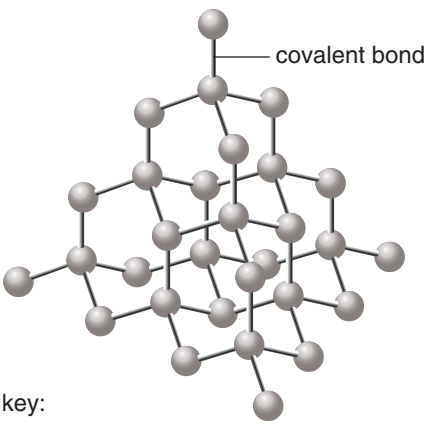


Fig. 41.7 Arrangement of atoms in silicon

Giant covalent structure

The metalloids (boron and silicon) and the non-metal carbon (diamond) have giant covalent structures. N1

Fig. 41.7 shows the arrangement of atoms in silicon. Each silicon atom is strongly bound by covalent bonds to four other silicon atoms. The covalent linking extends from one atom to the next throughout the whole lattice, forming a three-dimensional giant structure.

ASL 2006 Q5(a)(i)
ASL 2009 Paper 1 Q9
AL 2010 Paper 1 Q1(c)

delocalized electron 離域電子

Simple molecular structure

ASL 2009 Paper 1 Q9

All the non-metals in periods 2 and 3 (except carbon) form simple molecular structures. Each of these elements consists of separate small molecules, i.e. N_2, O_2, F_2, Ne, P_4, S_8, Cl_2, Ar.

There are strong covalent bonds within these molecules but only weak van der Waals' forces between separate molecules.

Neon and argon exist as monoatomic molecules and there are only van der Waals' forces between them.

Allotropes are two (or more) forms of the same element in which the atoms or molecules are arranged in different ways.

We have discussed allotropes in Topic 2 Microscopic World I.

Phosphorus has several allotropes, one of which is a waxy solid called white phosphorus (Fig. 41.8a). It is composed of P_4 molecules in which each phosphorus atom lies at a corner of a tetrahedron, as shown in Fig. 41.8b.



Fig. 41.8a White phosphorus sample

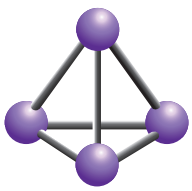
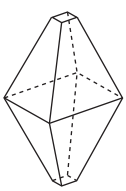


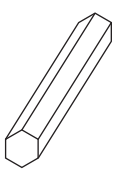
Fig. 41.8b The tetrahedral structure of white phosphorus

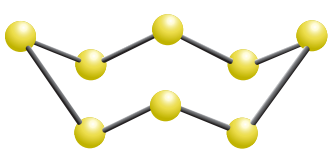
Sulphur also has several allotropes. *Rhombic sulphur* and *monoclinic sulphur* are the crystalline allotropes. Crystalline sulphur consists of *puckered* S_8 rings in the shape of crowns (Fig. 41.9).

Rhombic sulphur is stable below 96 °C. Above this temperature, monoclinic sulphur will form.

rhombic sulphur $\underset{\text{cool to below 96 °C}}{\overset{\text{heat to above 96 °C}}{\rightleftharpoons}}$ monoclinic sulphur







CE 2010 Paper 2 Q31

Fig. 41.9 The S_8 molecule

allotrope 同素異形體　rhombic sulphur 斜方晶硫　monoclinic sulphur 單斜晶硫　puckered 起褶子的

The S_8 molecules can be packed together in two different ways, forming rhombic crystals (Fig. 41.10a) and monoclinic crystals (Fig. 41.10b).



Fig. 41.10a Crystals of rhombic sulphur



Fig. 41.10b Crystals of monoclinic sulphur

41.4 Variation in atomic radius of elements across a period

Atomic radius

We have discussed this in Topic 2 Microscopic World I.

Atomic radius is generally stated as being the distance from an atom's nucleus to the boundary of the surrounding cloud of electrons. However, we do not know exactly where the boundary is. So we cannot measure the radius of an isolated atom. Instead, we measure the shortest distance between the nuclei of two identical touching atoms, and then halving that distance.

Because atoms combine with different types of bonding, we have different types of atomic radii.

- **Covalent radius** is half the shortest distance from one nucleus to the next in a covalently bonded element (Fig. 41.11).
- **Metallic radius** is half the shortest distance from the nucleus of one metallic ion to the next (Fig. 41.12).

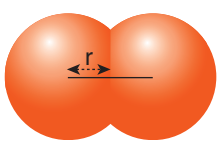


Fig. 41.11 Covalent radius

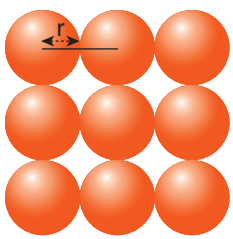


Fig. 41.12 Metallic radius

atomic radius 原子半徑　　covalent radius 共價半徑　　metallic radius 金屬半徑

Even metals can form covalent molecules in the gas state.

The covalent radius can be measured for most elements and is usually what we mean when we use the term 'atomic radius'.

Fig. 41.13 shows the variation in the atomic radius of the first 20 elements. The graph shows that

- atomic radius has a periodic property because it decreases across each period and there is a jump when we start the next period;
- atoms get larger as we move down a group.

1 pm (picometre) = 10^{-12} m

Metallic radii are used for Li, Be, Na, Mg, Al, K and Ca. Covalent radii are used for the other elements.

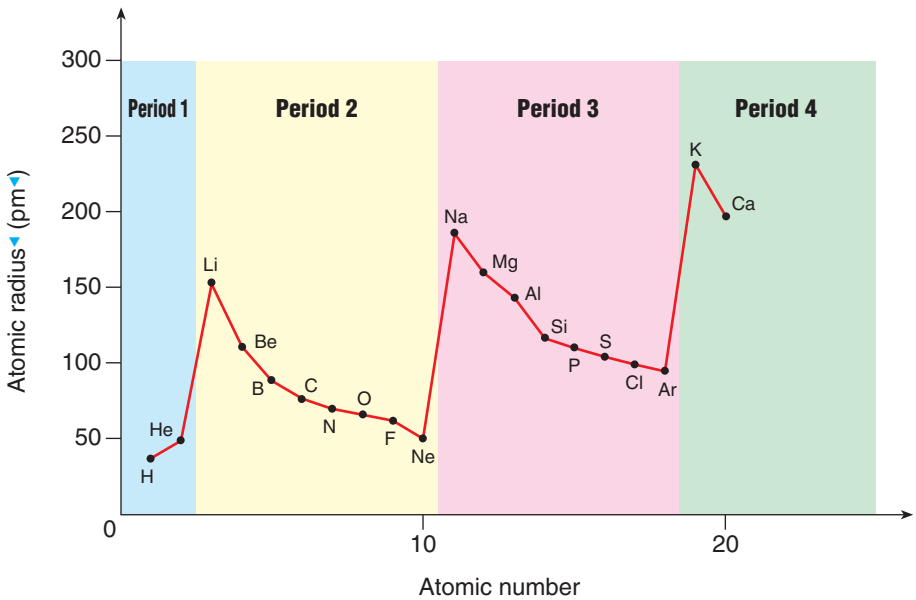


Fig. 41.13 Variation in atomic radius of the first 20 elements

Why do the radii of atoms decrease across a period?

All electrons are negatively charged and repel each other. Electrons in the inner shells tend to shield the outermost shell electrons from the attractive force of the nucleus. This is called the **shielding effect**.

For example, the electronic arrangement of a lithium atom is 2,1. The outermost shell electron is attracted by a nucleus with a charge of 3+ shielded by two electrons. The **effective nuclear charge** felt by the outermost shell electron is closer to 1+.

shielding effect 屏蔽效應　　effective nuclear charge 有效核電荷

ASL 2009 Paper 1 Q9

As we move across Period 3 (i.e. from sodium to argon), the numbers of protons and electrons both increase by one with each successive element. The charge on the nucleus increases from 11+ to 18+. The extra electron added to the atom of each successive element occupies the third electron shell. The shielding effect provided by the two inner electron shells remains roughly the same. So the effective nuclear charge felt by the outermost shell electrons increases as we move across the period. The electrons are pulled closer to the nucleus. Hence the atomic radius decreases across the period (Fig. 41.14). N2

N2 Refer to page T10.

Atom	Na	Mg	Al	Si	P	S	Cl	Ar
Electronic arrangement of atom	2,8,1	2,8,2	2,8,3	2,8,4	2,8,5	2,8,6	2,8,7	2,8,8
Nuclear charge	11+	12+	13+	14+	15+	16+	17+	18+
Atomic radius (pm)	186	160	143	117	110	104	99	94

Fig. 41.14 The atomic radii and electronic arrangements of atoms of Period 3 elements

Why do the radii of atoms increase down a group?

As we move down a group in the periodic table, the atom of each element has one extra full shell of electrons compared with the one before. So, although the nucleus also gains positive protons, the outermost shell electrons are both further away and shielded by more electron shells. As a result, these electrons are held less tightly and the atomic radius increases (Fig. 41.15).

Atom	Electronic arrangement of atom	Nuclear charge	Atomic radius (pm)
Li	2,1	3+	152
Na	2,8,1	11+	186
K	2,8,8,1	19+	227

Fig. 41.15 The atomic radii and electronic arrangements of atoms of the first three Group I elements

41.5 Trend in melting points of Period 3 elements

Physical properties, such as the melting point and electrical conductivity of elements, also show trends across a period. Fig. 41.16 shows the variation of melting points of Period 3 elements. The melting points increase from sodium to a maximum at Group IV's silicon and then decrease dramatically from silicon to argon.

CE 2008 Paper 2 Q22

ASL 2012 Paper 2 Q5
AL 2012 Paper 2 Q6

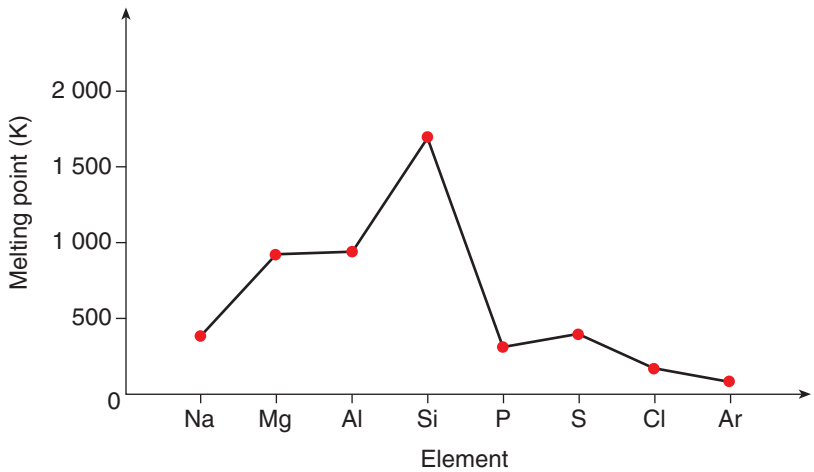


Fig. 41.16 A graph showing the variation of melting points of Period 3 elements

The melting point of an element depends on the strength of its inter-particle attractions. To explain the trend in melting points, we have to consider again the structures and bonding of the elements (Table 41.2).

Table 41.2

The structures, bonding and melting points of Period 3 elements

Group	I	II	III	IV	V	VI	VII	0
Element	Na	Mg	Al	Si	P (white)	S (rhombic)	Cl	Ar
Type of structure	giant metallic			giant covalent	simple molecular			
					P_4	S_8	Cl_2	Ar
Types of bonding and intermolecular forces	metallic			covalent	covalent bond holding atoms together within the molecule; van der Waals' forces between molecules			van der Waals' forces between monoatomic molecules
Melting point (K)	371	923	933	1 683	317	386	172	84

Sodium, magnesium and aluminium

Sodium, magnesium and aluminium are metals. They have metallic bonding, in which positive metallic ions are attracted to delocalized electrons. The melting point increases as the strength of the metallic bond increases.

The strength of metallic bond is affected by

- the number of outermost shell electrons available for metallic bonding per atom;
- the size of the positive metallic ion;
- the degree of compactness of the packing of the positive metallic ions.

The charge on the positive metallic ion increases as we pass along the first three metals of Period 3. Sodium has a charge of +1 while magnesium is +2 and aluminium is +3. There is a similar increase in the number of delocalized electrons available for bonding. The greater is the number of delocalized electrons available for bonding, the stronger are the metallic bonds.

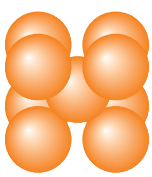
ASL 2009 Paper 1 Q9
ASL 2012 Paper 2 Q5
AL 2012 Paper 2 Q6

Furthermore, the atomic radius decreases from sodium to aluminium. The same trend is repeated in the ion. The smaller is the ion, the closer the electrons are to the positive nucleus and the stronger is the attraction.

In addition, the degree of compactness of the packing of positive metallic ions increases from sodium to aluminium.

Hence the strength of metallic bonds and melting points of the three metals increase in the order: sodium < magnesium < aluminium (Fig. 41.17).

Positive metallic ions in metals are packed in a very regular way. For example, those in sodium are packed as shown below. Each ion touches four ions in the layer above and four ions in the layer below.



On the other hand, positive metallic ions in magnesium and aluminium are packed more closely than those in sodium.

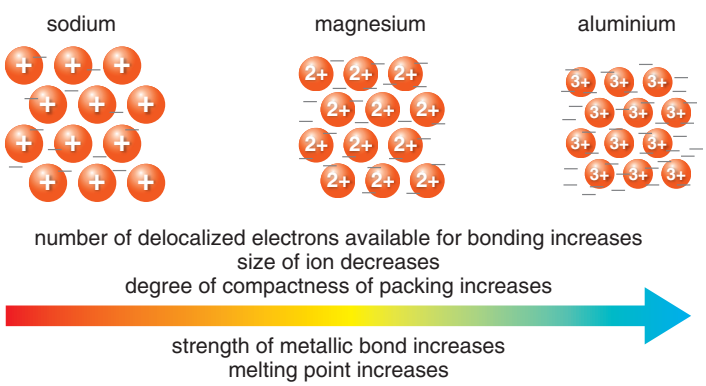


Fig. 41.17 The strength of metallic bonds and melting points of the three metals increase in the order: sodium < magnesium < aluminium

Silicon

DSE 2012 Paper 1B Q16(b)

Silicon has a giant covalent structure. To melt silicon, many of the covalent bonds between the atoms must be broken. This needs a lot of heat. Hence it has a very high melting point.

Non-metals

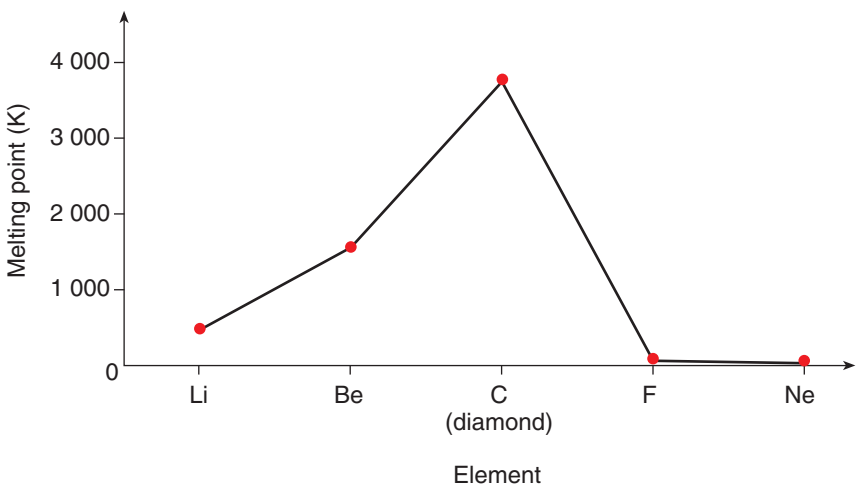
Phosphorus exists as P_4 molecules, sulphur as S_8 molecules, chlorine as Cl_2 molecules and argon as monoatomic molecules. Molecules of these non-metals are held together by weak van der Waals' forces. Only very little heat is needed to separate the molecules. Hence they have low melting points.

The strength of van der Waals' forces between molecules increases with the molecular size (or the number of electrons in one molecule). Hence the strength of van der Waals' forces is in the order: $S_8 > P_4 > Cl_2 > Ar$. So, the heat required to separate the molecules decreases in the same order, as shown by their decreasing melting points.

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

The following graph shows the variation in the melting points of five Period 2 elements. Account for the variation in terms of the type of structure and bonding.



DSE 2013 Paper 1B Q13

Discussion

Fluorine and neon have simple molecular structures. Weak van der Waals' forces exist between the molecules. Thus, they have a low melting points.

Lithium and beryllium have giant metallic structures. Metallic bonds exist in both of them.

Carbon (diamond) has a giant covalent structure. Strong covalent bonds exist between the atoms. Thus, it has the highest melting point among the five elements.

As compared with lithium,

- beryllium has a greater number of outermost shell electrons available for bonding;
- a beryllium ion has a smaller size;
- beryllium ions are more closely packed in the structure.

Thus, the strength of metallic bond and melting point of beryllium is higher than that of lithium.

Fluorine exists as F_2 molecules while neon exists as monoatomic molecules.

The size of a F_2 molecule is larger than a neon atom. / There are more electrons in a F_2 molecule than in a neon atom.

Hence the van der Waals' forces between fluorine molecules are stronger than those between neon atoms.

So, more heat is needed to separate the fluorine molecules during melting.

Variation in melting points of Group I elements

Fig. 41.18 shows the variation in the melting points of the first three Group I elements.



Fig. 41.18 The variation in the melting points of the first three Group I elements

For Group I metals, the number of outermost shell electrons available for bonding per atom is the same. The degree of compactness of the packing of the positive metallic ions is similar.

However, the size of the positive metallic ion increases as we move down the group. The delocalized electrons are further away from the positive nucleus. It becomes easier to break the metallic bonds and thus the melting point decreases as we move down the group.

Variation in melting points of Group 0 elements

Fig. 41.19 shows the variation in the melting points of the first three Group 0 elements.

ASL 2010 Paper 2 Q5(a)

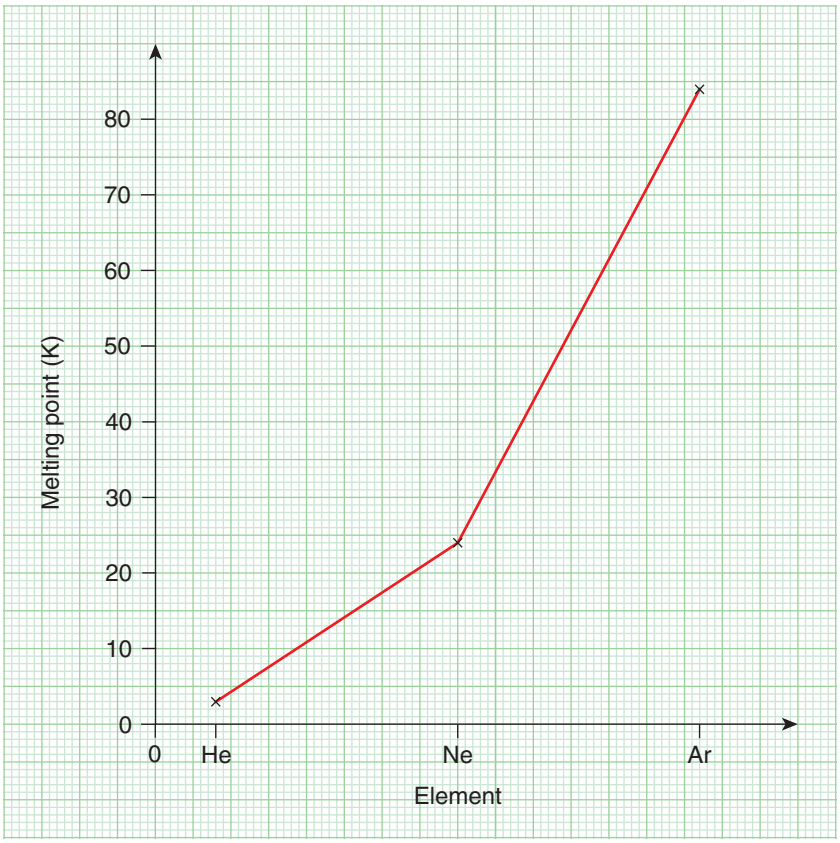


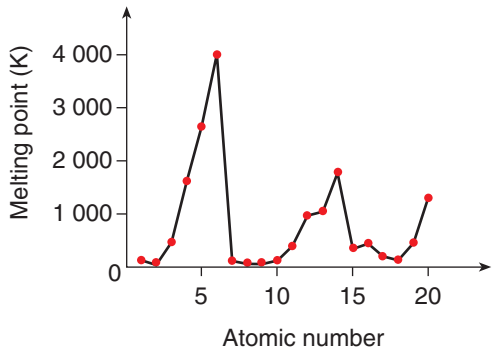
Fig. 41.19 The variation in the melting points of the first three Group 0 elements

The intermolecular attractions in Group 0 elements are van der Waals' forces. The atomic size (or the number of electrons in one atom) increases as we move down the group. The strength of van der Waals' forces and thus the melting points of the elements increase as we move down the group.

Practice 41.1

Refer to page T11.

1 The graph below shows the variation in melting points of the first 20 elements.



a) Which elements are at the peaks of the melting point graph? Which elements are in the troughs?

b) Explain why this graph provides evidence for periodicity.

2 The symbols r, s, t, u, v, w, x and y represent eight consecutive elements in the third and fourth periods of the periodic table. The table below lists their melting points:

Element	r	s	t	u	v	w	x	y
Melting point (K)	933	1 683	317	386	172	84	337	1 111

a) Deduce from the above information which elements s and t represent respectively.

b) Explain why the melting point of u is higher than that of v.

c) Explain why the melting point of x is lower than that of y.

ASL 2006 Q11(b)

Known values of electrical conductivities are shown here. Electrical conductivities are measured in units called 'siemens per metre' (S m^{-1}). Siemens are the reciprocal of the units of electrical resistance (ohms, Ω), i.e. S = Ω^{-1}.

41.6 Variation in electrical conductivities of Period 3 elements

Table 41.3 lists the **electrical conductivities** of Period 3 elements. The electrical conductivity increases across Period 3 from sodium to aluminium, then decreases to silicon, which is a semiconductor. Phosphorus and sulphur have negligible electrical conductivities. They are both insulators.

Table 41.3

The electrical conductivities of Period 3 elements

Group	I	II	III	IV	V	VI	VII	0
Element	Na	Mg	Al	Si	P (white)	S (rhombic)	Cl	Ar
Type of element	metals			metalloid	non-metals			
Electrical conductivity (x 10^8 S m^{-1})	0.218	0.224	0.382	2 x 10^{-10}	10^{-17}	10^{-23}	—	—

electrical conductivity 導電性

DSE 2014 Paper 1A Q30

Fig. 41.20 shows the relative electrical conductivities of Period 3 elements.

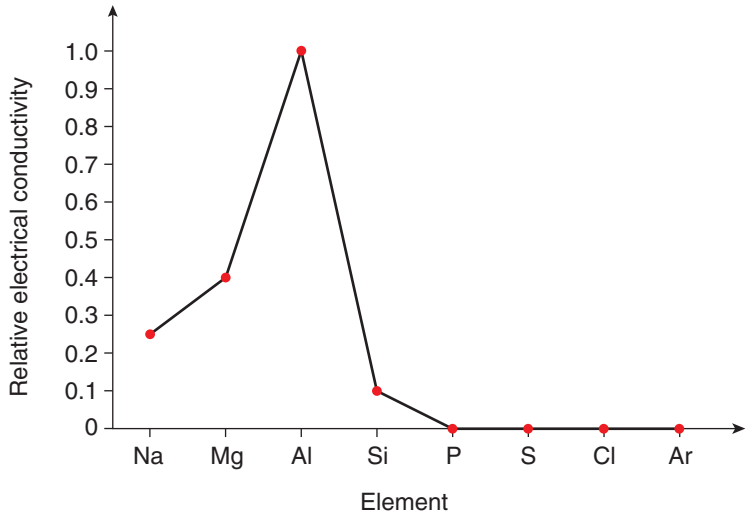


Fig. 41.20 Relative electrical conductivities of Period 3 elements

Sodium, magnesium and aluminium

ASL 2012 Paper 1 Q11
AL 2005 Paper 1 Q1(b)(i),(ii)
AL 2012 Paper 1 Q9

Sodium, magnesium and aluminium are all metals. They have metallic bonding. They are good conductors of electricity due to the movement of the mobile electrons. When a potential difference is applied, these electrons drift through the metal and the connecting wires towards the positive terminal.

Going from sodium to aluminium, the number of delocalized electrons increases. There are more electrons available to drift through the structure when the metal conducts a current. So, the electrical conductivity increases.

Silicon

In silicon, the four outermost shell electrons in each silicon atom are held in the four covalent bonds which link the atom to four other silicon atoms.

At room temperature, very few electrons gain enough energy to become delocalized. So, there are few delocalized electrons and silicon is a poor conductor.

At higher temperatures, more electrons gain enough energy and become delocalized. Hence there are more mobile electrons, which enable the conduction of electricity. This is why silicon is referred to as a semiconductor.

Non-metals

The other elements in Period 3 from phosphorus to argon do not conduct electricity at all. This is because they do not contain mobile electrons.

Table 41.4 summarizes the variation in the structures, bonding and physical properties of elements across periods 2 and 3.

Table 41.4

The variation in the structures, bonding and physical properties of elements across periods 2 and 3

Group	I	II	III	IV	V	VI	VII	0
Element in Period 2	Li	Be	B	C (diamond)	N	O	F	Ne
Type of element	metals		metalloid	non-metals				
Type of structure	giant metallic		giant covalent		simple molecular			
Types of bonding and intermolecular forces	metallic		covalent		covalent bond holding atoms together within the molecule; van der Waals' forces between molecules			van der Waals' forces between monoatomic molecules
Melting point (K)	453	1 553	2 349	3 773	63	54	53	24
Electrical conductivity	good conductors		semiconductor	non-conductors				
Element in Period 3	Na	Mg	Al	Si	P (white)	S (rhombic)	Cl	Ar
Type of element	metals			metalloid	non-metals			
Type of structure	giant metallic			giant covalent	simple molecular			
Types of bonding and intermolecular forces	metallic			covalent	covalent bond holding atoms together within the molecule; van der Waals' forces between molecules			van der Waals' forces between monoatomic molecules
Melting point (K)	371	923	933	1 683	317	386	172	84
Electrical conductivity	good conductors			semiconductor	non-conductors			

✓ To summarize, the main trends across a period of elements:

- a change from metals through metalloids to non-metals; N3 N4
- a decrease in atomic radius;
- a maximum in melting point is shown in the middle of the period;
- a change from conductors to non-conductors.

N3 Refer to page T11.

N4 Refer to page T12.

41.7 Reactions of Period 3 elements with oxygen N5

N5 Refer to page T12.

Sodium, magnesium and aluminium

Sodium reacts vigorously when heated and placed in a gas jar of oxygen. The sodium burns with a yellow flame (Fig. 41.21a). The main product is a white solid, sodium oxide (Fig. 41.21b).

$$4Na(s) + O_2(g) \longrightarrow 2Na_2O(s)$$

Sodium peroxide (Na_2O_2) is also formed.

$$2Na(s) + O_2(g) \longrightarrow Na_2O_2(s)$$



Fig. 41.21a Sodium burning in oxygen



Fig. 41.21b Sodium oxide is a white solid

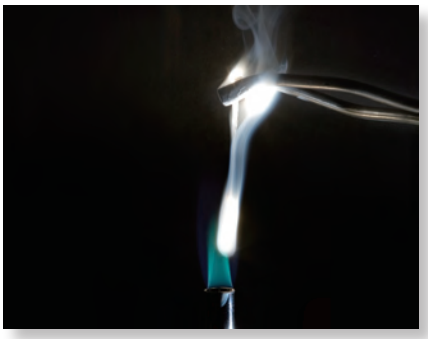


Fig. 41.22 Magnesium burns with a bright white flame

Magnesium also reacts vigorously when heated, forming magnesium oxide (Fig. 41.22). Aluminium metal is protected by a tough layer of aluminium oxide, but powdered aluminium reacts well with oxygen. Both metals burn with bright white flames.

$$2Mg(s) + O_2(g) \longrightarrow 2MgO(s)$$

$$4Al(s) + 3O_2(g) \longrightarrow 2Al_2O_3(s)$$

Silicon

Silicon reacts slowly with oxygen to form silicon dioxide.

$$Si(s) + O_2(g) \longrightarrow SiO_2(s)$$

Phosphorus

White phosphorus reacts vigorously with oxygen. A white flame is seen and clouds of white phosphorus pentoxide (P_4O_{10}) are produced.

$$P_4(s) + 5O_2(g) \longrightarrow P_4O_{10}(s)$$



Fig. 41.23 Sulphur burns in oxygen with a blue flame to give sulphur dioxide gas

Sulphur

Sulphur powder burns gently with a blue flame in oxygen. Toxic sulphur dioxide gas is produced (Fig. 41.23).

$$S(s) + O_2(g) \longrightarrow SO_2(g)$$

Further oxidation of sulphur dioxide gives sulphur trioxide.

$$2SO_2(g) + O_2(g) \underset{}{\overset{V_2O_5 \text{ catalyst}}{\rightleftharpoons}} 2SO_3(g)$$

Chlorine

Despite having several oxides, chlorine does not react directly with oxygen.

41.8 Properties of oxides of Period 3 elements

Table 41.5 lists some properties of oxides of Period 3 elements. These oxides can be grouped into three types:

- **ionic oxides**;
- **covalent oxide** with a giant structure;
- covalent oxides with simple molecular structures.

The ability of a substance to conduct electricity when it is in a liquid state is an indication of the ionic character of the substance. The molten oxides of sodium, magnesium and aluminium show good electrical conductivity, indicating that these are ionic oxides.

DSE 2012 Paper 1B Q16(a)
AL 2012 Paper 1 Q9

Table 41.5

Some properties of oxides of Period 3 elements

Element	Na	Mg	Al	Si	P	S	Cl
Formula(e) of oxide(s)	Na_2O	MgO	Al_2O_3	SiO_2	P_4O_{10}	SO_2 SO_3	Cl_2O
Relative melting point	high	very high			low		
Electrical conductivity when in liquid state	good			none			
Type of structure	giant ionic			giant covalent	simple molecular		
Type of bonding	ionic	mainly ionic		mainly covalent			

AL 2006 Paper 1 Q1(a)

DSE 2013 Paper 1A Q36

ionic oxide 離子氧化物　　covalent oxide 共價氧化物

41.9 Composition of the oxides of Period 3 elements

Table 41.6 summarizes the formulae of oxides of periods 2 and 3 elements.

Table 41.6

The formulae of oxides of periods 2 and 3 elements							
Group	I	II	III	IV	V	VI	VII
Element in Period 2	Li	Be	B	C (diamond)	N	O	F
Formula(e) of oxide(s)	Li_2O	BeO	B_2O_3	CO CO_2	N_2O NO N_2O_3 NO_2 N_2O_4 N_2O_5	O_2	F_2O
Element in Period 3	Na	Mg	Al	Si	P	S	Cl
Formula(e) of oxide(s)	Na_2O	MgO	Al_2O_3	SiO_2	P_4O_6 P_4O_{10}	SO_2 SO_3	Cl_2O ClO_2 Cl_2O_6 Cl_2O_7

For a non-metal that forms more than one oxide, the oxide shown here is one with the element in its highest oxidation state.

Fig. 41.24 shows the number of moles of oxygen atoms combining with one mole of atoms of each element in periods 2 and 3. For example, Al_2O_3 corresponds to 1.5 moles of oxygen atoms per mole of aluminium atoms.

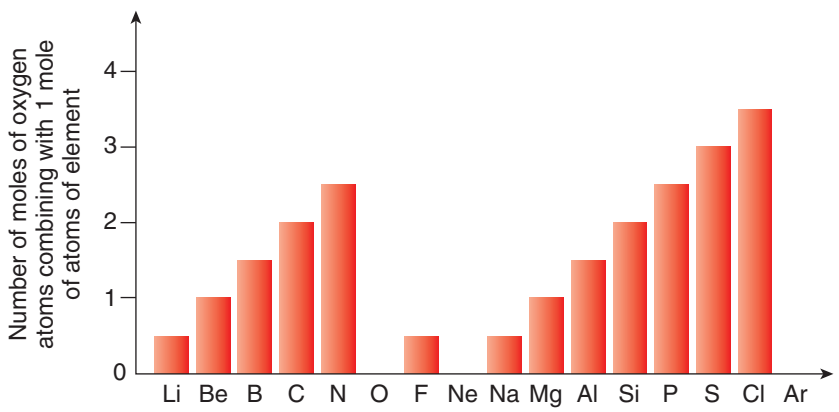


Fig. 41.24 Number of moles of oxygen atoms combining with one mole of atoms of each element in periods 2 and 3

Going across periods 2 and 3, there is a regular increase in the number of moles of oxygen atoms that combines with one mole of atoms of each element. This number reaches a maximum at Group V or Group VII.

41.10 Effect of water on oxides of Period 3 elements

41.1
Investigating the acid-base properties of oxides of Period 3 elements.

Ionic oxides

ASL 2007 Paper 2 Q2(c)

Sodium oxide reacts with water to form sodium hydroxide. The presence of excess hydroxide ions makes the solution alkaline. N6

$$Na_2O(s) + H_2O(l) \longrightarrow 2NaOH(aq)$$

strongly alkaline solution

Sodium peroxide reacts with water to form sodium hydroxide solution and hydrogen peroxide solution. N7

$$Na_2O_2(s) + 2H_2O(l) \longrightarrow 2NaOH(aq) + H_2O_2(aq)$$

ASL 2008 Paper 2 Q2(b)(ii)
AL 2008 Paper 2 Q1(b)(ii)

Magnesium oxide reacts slightly with cold water and moderately with hot water to form magnesium hydroxide, which is very slightly soluble. So, only a slightly alkaline solution is formed. N8

$$MgO(s) + H_2O(l) \xrightarrow{\text{heat}} Mg(OH)_2(s)$$

DSE 2013 Paper 1A Q36

Aluminium oxide does not react with water. However, it reacts with an acid or an alkali.

DSE 2014 Paper 1A Q36

ASL 2010 Paper 2 Q5(c)

- With acid:

  e.g. $Al_2O_3(s) + 3H_2SO_4(aq) \longrightarrow Al_2(SO_4)_3(aq) + 3H_2O(l)$

  or

  $Al_2O_3(s) + 6H^+(aq) \longrightarrow 2Al^{3+}(aq) + 3H_2O(l)$

- With hot concentrated alkali:

  e.g. $Al_2O_3(s) + 2NaOH(aq) + 3H_2O(l) \longrightarrow 2NaAl(OH)_4(aq)$

  DSE 2012 Paper 1B Q16(c)

  or

  $Al_2O_3(s) + 2OH^-(aq) + 3H_2O(l) \longrightarrow 2[Al(OH)_4]^-(aq)$

N6 Sodium oxide also reacts with acids.
e.g. $Na_2O(s) + 2HCl(aq) \longrightarrow 2NaCl(aq) + H_2O(l)$

N7 Superoxides react with acids to form salts, hydrogen peroxide solution and oxygen.
e.g. $2KO_2(s) + 2HCl(aq) \longrightarrow 2KCl(aq) + H_2O_2(aq) + O_2(g)$

N8 Magnesium oxide also reacts with acids.
e.g. $MgO(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2O(l)$

DSE 2013 Paper 1A Q36
DSE 2014 Paper 1A Q36

When aluminium oxide reacts with an acid, it behaves like a base — forming a salt plus water. On the other hand, when the oxide reacts with an alkali, it behaves like an acid — forming a salt. Compounds which can act as both acids and bases, such as aluminium oxide, are called **amphoteric**. N9 N10

N9 Examination questions often ask about the amphoteric property of aluminium oxide.

N10 Beryllium oxide is also amphoteric. It does not dissolve nor react with water, but reacts with both acids and bases.

$BeO(s) + 2H^+(aq) \longrightarrow Be^{2+}(aq) + H_2O(l)$

$BeO(s) + 2OH^-(aq) + H_2O(l) \longrightarrow [Be(OH)_4]^{2-}(aq)$

beryllate ion

Amphoteric nature is due to the nature of the metal-oxygen bond which is partially ionic and partially covalent, giving an intermediate character.

ASL 2007 Paper 2 Q2(c)

Covalent oxide with a giant structure

Silicon dioxide is insoluble in water. However, it reacts with hot concentrated alkali.

$$SiO_2(s) + 2NaOH(aq) \xrightarrow{\text{heat}} Na_2SiO_3(aq) + H_2O(l)$$

Silicon dioxide acts as an acid, forming a salt plus water with an alkali. So, it is classified as an **acidic oxide**. N11

N11 **Covalent oxides of Period 2 elements**

Boron trioxide reacts with water to form boric acid.

$B_2O_3(s) + 3H_2O(l) \longrightarrow 2H_3BO_3(aq)$

Carbon monoxide is neutral and insoluble in water.

Carbon dioxide is moderately soluble in cold water, forming a slightly acidic solution.

$CO_2(g) + H_2O(l) \rightleftharpoons H_2CO_3(aq)$

Both dinitrogen monoxide and nitrogen monoxide are neutral and insoluble in water.

Nitrogen dioxide reacts rapidly with cold water to form nitrous acid and nitric acid.

$2NO_2(g) + H_2O(l) \longrightarrow HNO_2(aq) + HNO_3(aq)$

Oxygen is neutral and very slightly soluble in water.

Oxygen difluoride is a colourless gas which hydrolyzes slowly in water to form oxygen and hydrogen fluoride.

$F_2O(g) + H_2O(l) \longrightarrow O_2(g) + 2HF(aq)$

Covalent oxides with simple molecular structures

Phosphorus pentoxide reacts vigorously with cold water to form metaphosphoric acid.

$$P_4O_{10}(s) + 2H_2O(l) \longrightarrow 4HPO_3(aq)$$

metaphosphoric acid

Phosphorus pentoxide also reacts with warm water to form phosphoric acid.

$$P_4O_{10}(s) + 6H_2O(l) \longrightarrow 4H_3PO_4(aq)$$

phosphoric acid

ASL 2007 Paper 2 Q2(c)

Sulphur dioxide and sulphur trioxide react with water to form sulphurous acid and sulphuric acid respectively. N12

$$SO_2(g) + H_2O(l) \rightleftharpoons H_2SO_3(aq)$$

sulphurous acid

$$SO_3(g) + H_2O(l) \rightleftharpoons H_2SO_4(aq)$$

sulphuric acid

Dichlorine monoxide dissolves in water to form hypochlorous acid.

$$Cl_2O(g) + H_2O(l) \longrightarrow 2HOCl(aq)$$

hypochlorous acid

ASL 2008 Paper 2 Q2(b)(iii)
AL 2008 Paper 2 Q1(b)(iii)

N12 Sulphur dioxide also reacts with bases. If $SO_2(g)$ is bubbled through sodium hydroxide solution, sodium sulphite solution forms, followed by sodium hydrogensulphite solution when $SO_2(g)$ is in excess.

$SO_2(g) + 2NaOH(aq) \longrightarrow Na_2SO_3(aq) + H_2O(l)$

$Na_2SO_3(aq) + H_2O(l) + SO_2(g) \longrightarrow 2NaHSO_3(aq)$

amphoteric 兩性的　　acidic oxide 酸性氧化物

Neutral oxides show no acidic and basic properties. Carbon monoxide and nitrogen monoxide are examples.

Table 41.7 lists the acid-base nature of oxides of Period 3 elements. In general, metals react with oxygen to form ionic oxides which are basic. Most covalent oxides formed from non-metals are acidic, but some are neutral. Aluminium oxide is an **amphoteric oxide**.

Table 41.7

The acid-base nature of oxides of Period 3 elements N13

Element	Na	Mg	Al	Si	P	S	Cl
Formula(e) of oxide(s)	Na_2O	MgO	Al_2O_3	SiO_2	P_4O_{10}	SO_2 SO_3	Cl_2O
Type of structure	giant ionic			giant covalent	simple molecular		
Type of bonding	ionic	mainly ionic		mainly covalent			
Acid-base nature	strongly basic	basic	amphoteric	acidic			strongly acidic

ASL 2008 Paper 2 Q2(b)(i)
AL 2008 Paper 2 Q1(b)(i)

N13 Refer to page T13.

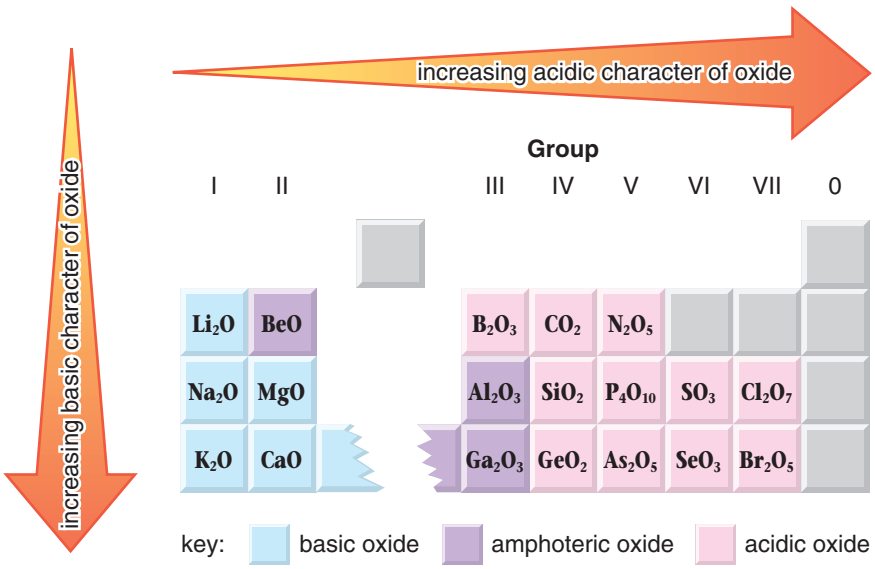
Summary of acid-base nature of oxides of periods 2, 3 and 4 elements

Fig. 41.25 shows the formulae and acid-base nature of oxides of periods 2, 3 and 4 elements. We can see certain trends emerging:

- Ionic oxides of metals (on the left of the periodic table) have giant ionic structures and react with water to give alkalis. They are **basic oxides**.
- Covalent oxides of non-metals (on the right of the periodic table) have simple molecular structures and react with water to give acids. They are acidic oxides.
- Oxides of elements in the middle of the periodic table have intermediate properties. They are amphoteric oxides.

ASL 2007 Paper 2 Q2(c)

amphoteric oxide 兩性氧化物　　basic oxide 鹼性氧化物



For a non-metal that forms more than one oxide, the oxide shown here is one with the element in its highest oxidation state.

Fig. 41.25 Acid-base nature of oxides of periods 2, 3 and 4 elements

Example 41.1

Q Consider the following oxides of Period 3 elements:

Na_2O MgO Al_2O_3 SiO_2 P_4O_{10} SO_2

a) Which of the oxides listed above can conduct electricity in molten state?

b) i) Using the following notations, complete the information about the acid-base nature of the oxides.

AC: acidic

BA: basic

AM: amphoteric

Oxide	Na_2O	MgO	Al_2O_3	SiO_2	P_4O_{10}	SO_2
Acid-base nature						

ii) Outline chemical tests to show how these oxides can be classified into acidic, basic and amphoteric.

Continued on next page

A1 a) Na_2O, MgO and Al_2O_3

b) i)

Oxide	Na_2O	MgO	Al_2O_3	SiO_2	P_4O_{10}	SO_2
Acid-base nature	BA	BA	AM	AC	AC	AC

ii) Add each oxide to HCl(aq) and measure the pH of the mixture. Only Na_2O, MgO and Al_2O_3 react with HCl(aq) and the pH increases. These oxides demonstrate basic properties.

Add each oxide to NaOH(aq) and measure the pH of the mixture. Only Al_2O_3, SiO_2, P_4O_{10} and SO_2 react with NaOH(aq) and the pH decreases. These oxides demonstrate acidic properties.

When Al_2O_3 reacts with HCl(aq), it behaves like a base. When it reacts with NaOH(aq), it behaves like an acid. Thus, Al_2O_3 is amphoteric.

Practice 41.2

1 Germanium is a Group IV element in Period 4. It is a metalloid, as is silicon in Period 3.

a) Predict the type of bonding and structure of germanium. Type of bonding: covalent bonding
Type of structure: giant covalent structure

b) Germanium dioxide has properties similar to those of silicon dioxide. It is an acidic oxide.

Write a chemical equation for the reaction of germanium dioxide with hot concentrated sodium hydroxide solution.
$GeO_2(s) + 2NaOH(aq) \longrightarrow Na_2GeO_3(aq) + H_2O(l)$

2 Potassium oxide (K_2O) is a basic oxide. It reacts with water to form an alkaline solution.

a) Predict the type of bonding and structure of potassium oxide. Type of bonding: ionic bonding
Type of structure: giant ionic structure

b) Write a chemical equation for each of the following reactions of potassium oxide:

i) with water;
$K_2O(s) + H_2O(l) \longrightarrow 2KOH(aq)$

ii) with dilute nitric acid.
$K_2O(s) + 2HNO_3(aq) \longrightarrow 2KNO_3(aq) + H_2O(l)$

Have you mastered?

Key terms

main group element 主族元素 4 transition metal 過渡金屬 4 lanthanide 鑭系元素 4

actinide 錒系元素 4 periodicity 週期律 5 giant metallic structure 巨型金屬結構 6

giant covalent structure 巨型共價結構 6 simple molecular structure 簡單分子結構 6

van der Waals' forces 范德華力 6 delocalized electron 離域電子 7 allotrope 同素異形體 8

atomic radius 原子半徑 9 covalent radius 共價半徑 9 metallic radius 金屬半徑 9

shielding effect 屏蔽效應 10 effective nuclear charge 有效核電荷 10

electrical conductivity 導電性 17 ionic oxide 離子氧化物 22 covalent oxide 共價氧化物 22

amphoteric 兩性的 25 acidic oxide 酸性氧化物 25 amphoteric oxide 兩性氧化物 26

basic oxide 鹼性氧化物 26

Checklist

After studying this unit, you should be able to

- ☐ describe the type of structure and bonding of elements of Group I through Group 0 of the periodic table;
- ☐ describe the periodic variations of atomic radius, melting point and electrical conductivity of the elements;
- ☐ interpret the variations in melting point and in electrical conductivity in terms of the bonding and structure of the elements;
- ☐ describe the type of bonding and stoichiometric composition of the oxides of elements from Na to Cl;
- ☐ describe the variation in behaviour of the following oxides in water: Na_2O, MgO, Al_2O_3, SiO_2, P_4O_{10}, SO_2, SO_3 and Cl_2O;
- ☐ recognize the variation of the acid-base nature of the oxides of elements from Na to Cl.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 The following table summarizes the variation in the structures, bonding and physical properties of elements across periods 2 and 3.

Group	I	II	III	IV	V	VI	VII	0
Element in Period 2	Li	Be	B	C (diamond)	N	O	F	Ne
Type of element	metals		metalloid	non-metals				
Type of structure	giant metallic		giant covalent		simple molecular			
					N_2	O_2	F_2	Ne
Types of bonding and intermolecular forces	metallic		covalent		covalent bond holding atoms together within the molecule; van der Waals' forces between molecules			van der Waals' forces between monoatomic molecules
Melting point (K)	453	1 553	2 349	3 773	63	54	53	24
Electrical conductivity	good conductors		semiconductor	non-conductors				
Atomic radius	decrease →							
Element in Period 3	Na	Mg	Al	Si	P (white)	S (rhombic)	Cl	Ar
Type of element	metals			metalloid	non-metals			
Type of structure	giant metallic			giant covalent	simple molecular			
					P_4	S_8	Cl_2	Ar
Types of bonding and intermolecular forces	metallic			covalent	covalent bond holding atoms together within the molecule; van der Waals' forces between molecules			van der Waals' forces between monoatomic molecules
Melting point (K)	371	923	933	1 683	317	386	172	84
Electrical conductivity	good conductors			semiconductor	non-conductors			
Atomic radius	decrease →							

Have you mastered?

2 The main trends across a period of elements:

- a change from metals through metalloid to non-metals;
- a decrease in atomic radius;
- a maximum in melting point is shown in the middle of the period;
- a change from conductors to non-conductors.

3 For the Period 3 metals, from sodium to aluminium,

- the number of outermost shell electrons available for metallic bonding increases;
- the size of the positive metallic ion decreases;
- the degree of compactness of the packing of positive metallic ions in the structure increases.

Thus, the strength of metallic bonds and melting points increase from sodium to aluminium.

4 For the Period 3 non-metals, the molecular size (or the number of electrons in one molecule) is in the order:

$S_8 > P_4 > Cl_2 > Ar$.

Thus, the strength of van der Waals' forces and melting points are in the same order.

5 The following table summarizes the structure, bonding and acid-base nature of oxides of Period 3 elements.

Element	Na	Mg	Al	Si	P	S	Cl
Formula(e) of oxide(s)	Na_2O	MgO	Al_2O_3	SiO_2	P_4O_6 P_4O_{10}	SO_2 SO_3	Cl_2O
Type of structure	giant ionic			giant covalent	simple molecular		
Type of bonding	ionic	mainly ionic		mainly covalent			
Acid-base nature	strongly basic	basic	amphoteric	acidic			strongly acidic

6 Aluminium oxide is amphoteric. It can react with both acid and alkali.

e.g. $Al_2O_3(s) + 3H_2SO_4(aq) \longrightarrow Al_2(SO_4)_3(aq) + 3H_2O(l)$

$Al_2O_3(s) + 2NaOH(aq) + 3H_2O(l) \longrightarrow 2NaAl(OH)_4(aq)$

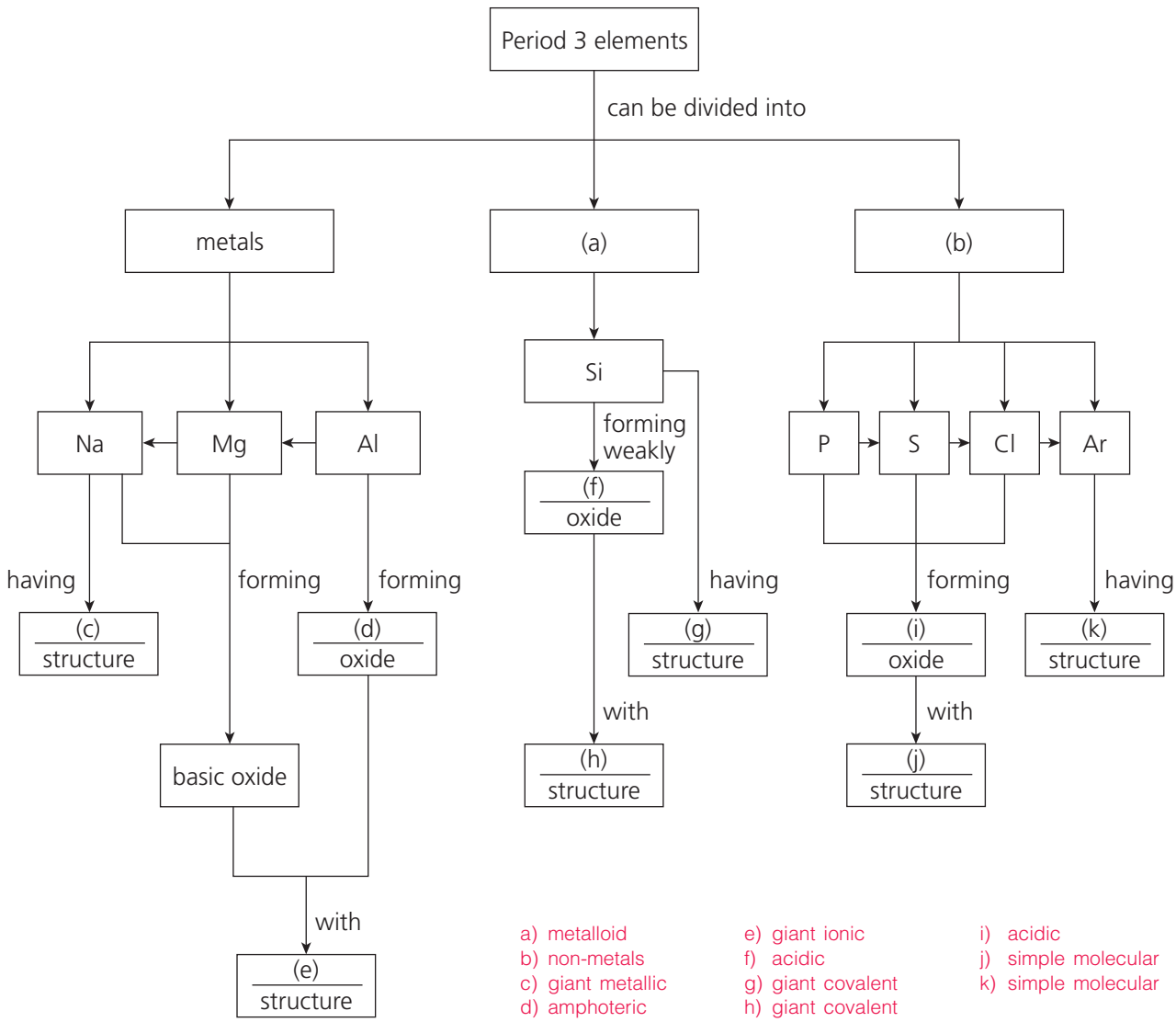
Have you mastered?

Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the following concept map.



a) metalloid
b) non-metals
c) giant metallic
d) amphoteric
e) giant ionic
f) acidic
g) giant covalent
h) giant covalent
i) acidic
j) simple molecular
k) simple molecular

6 The melting points of Period 3 elements increase from sodium to a maximum at silicon and then decrease from silicon to argon.

7 Going from sodium to aluminium, the number of delocalized electrons increases. There are more electrons available to drift through the structure when the metal conducts a current. So, the electrical conductivity increases from sodium to aluminium.

2 The elements in the third period of the periodic table are as follows:

Na Mg Al Si P S Cl Ar

Which of these elements

a) exists as diatomic molecules at room temperature and pressure?
Cl

b) has the highest melting point?
Si

c) has the highest electrical conductivity?
Al

d) has the smallest atomic radius?
Ar

3 a) When water was added to magnesium oxide, a solution of pH 11 was formed. When sulphur dioxide was bubbled into water, a solution of pH 2 was formed. Write an equation for each reaction and explain why each solution is NOT neutral.

b) Explain why silicon dioxide is INSOLUBLE in water.

3 a) $MgO(s) + H_2O(l) \longrightarrow Mg(OH)_2(s)$
Magnesium oxide reacts slightly with cold water to form magnesium hydroxide which is very slightly soluble. So, a slightly alkaline solution is formed.
$SO_2(g) + H_2O(l) \rightleftharpoons H_2SO_3(aq)$
Sulphur dioxide reacts with water to form sulphurous acid.

b) Silicon dioxide has a giant covalent structure.
The atoms are held together by strong covalent bonds and it is difficult to separate the atoms.

Part II Multiple choice questions

4 Which of the following elements has the SMALLEST atomic radius?

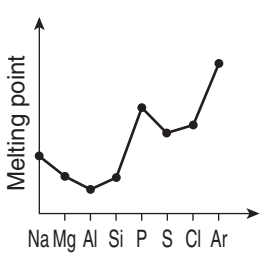
A C
(B) F The radii of atoms decrease across a period.
C N
D O

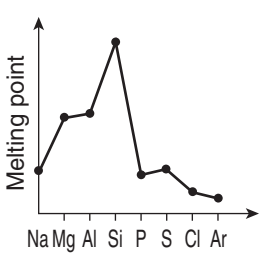
5 In which of the following series does the atomic radius of the element increase from left to right?

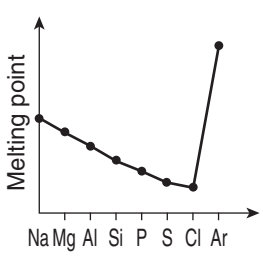
(A) Cl, Si, Na
B Ca, S, Mg
C K, Na, Ca
D Si, Mg, S

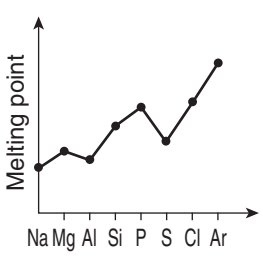
Cl, Si and Na belong to the same period. The atomic radii of elements decrease across a period. Thus, the atomic radius of the element increases in the order: Cl < Si < Na.

6 Which of the following graphs shows the variation of the melting point of Period 3 elements?
Refer to the previous page for explanation.

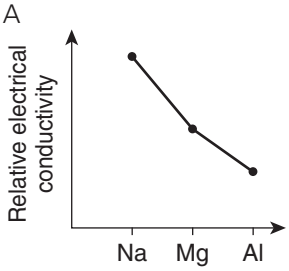
A


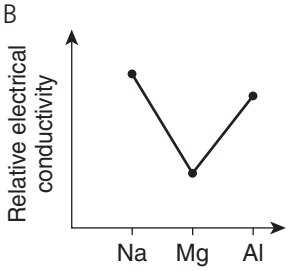
(B)


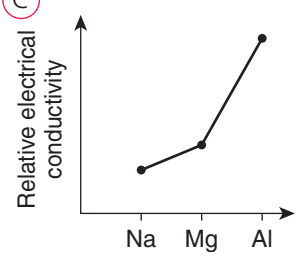
C


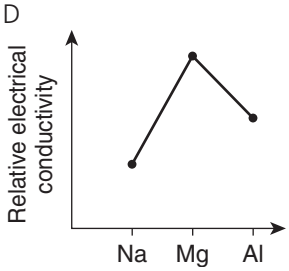
D


7 Which of the following graphs shows the variation of relative electrical conductivity of the three elements, sodium, magnesium and aluminium?
Refer to the previous page for explanation.

A


B


(C)


D


8 Which of the following oxides has a molecular structure?

A Na_2O
B Al_2O_3
C SiO_2
(D) Cl_2O_7

Chlorine is a non-metal. It forms oxides with molecular structures.

9 Which of the following oxides is basic?

A Al_2O_3
B GeO_2
C K_2O
D SO_3

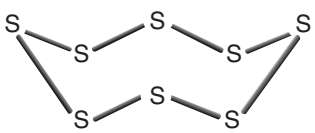
In general, metals react with oxygen to form ionic oxides which are basic.

10 X, Y and Z are three elements in the same short period of the periodic table. The oxide of X is acidic, the oxide of Y is basic and the oxide of Z is amphoteric. What is the order of increasing atomic number of these elements?

A X < Y < Z
B X < Z < Y
C Y < Z < X
D Z < Y < X

In general, metals react with oxygen to form basic oxides. Non-metals form acidic oxides.
Thus, X should be a non-metal while Y is a metal. Z is in between X and Y.
The order of increasing atomic number of the elements is: Y < Z < X.

11 The structure of a sulphur molecule in sulphur powder is shown below:



Which of the following statements is correct?

(Relative atomic mass: S = 32.1)

A The relative molecular mass of sulphur is 32.1.
B The oxidation number of sulphur in the molecule is 0.
C The attraction between sulphur molecules is covalent bond.
D Double bonds are present between adjacent atoms in sulphur molecule.

(HKCEE, Paper 2, 2010, 31)

12 Fluorine does NOT show variable oxidation state in its compounds. Its oxidation state is always –1.

12 Which of the following statements about the elements in Group VII is INCORRECT?

A They all show variable oxidation states in their compounds.
B They all form acidic hydrides.
C Electronegativity decreases as the group is descended.
D They all exist as diatomic molecules.

(Edexcel Advanced Subsidiary GCE, Unit 2, Jun. 2010, 11)

13 Which of the following oxides have giant structures?

(1) MgO
(2) P_4O_{10}
(3) SiO_2

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(2) P_4O_{10} has a simple molecular structure.

14 Which of the following trends occur(s) when going across the third period from sodium to aluminium?

(1) The melting point increases.
(2) The atomic radius increases.
(3) The bonding in the element changes from metallic to covalent.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

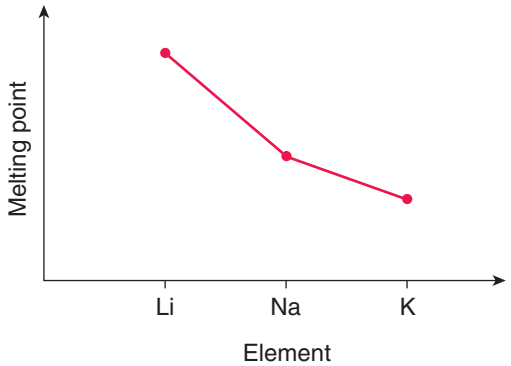
(2) The atomic radius decreases from sodium to aluminium.
(3) Metallic bonding exists in all the three metals.

Part III Structured questions

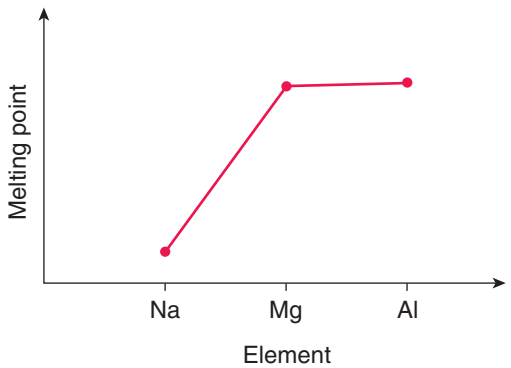
Answers for the questions of the public examinations in Hong Kong are not provided.

15 a) Sketch the trend in melting point of the first three Group I elements. Account for the trend.

Refer to page T13.



b) Sketch the trend in melting point of the Period 3 elements, Na, Mg and Al. Account for the trend.



16 The following table lists some information of three metals. Refer to page T14.

Metal	Relative atomic mass	Atomic radius (nm)
Mg	24.3	0.160
Al	27.0	0.143
Ca	40.1	0.197

Explain the order of atomic radius of the metals:

a) Mg > Al;

b) Mg < Ca.

17 The atomic radius of sodium is larger than that of chlorine.
A sodium atom has less protons than a chlorine atom.
The shielding effect provided by the two inner electron shells is the same for sodium and chlorine.
Thus, there is less attraction between the nucleus and the outermost shell electron of a sodium atom. The electron is held less tightly and the atomic radius of sodium is larger.

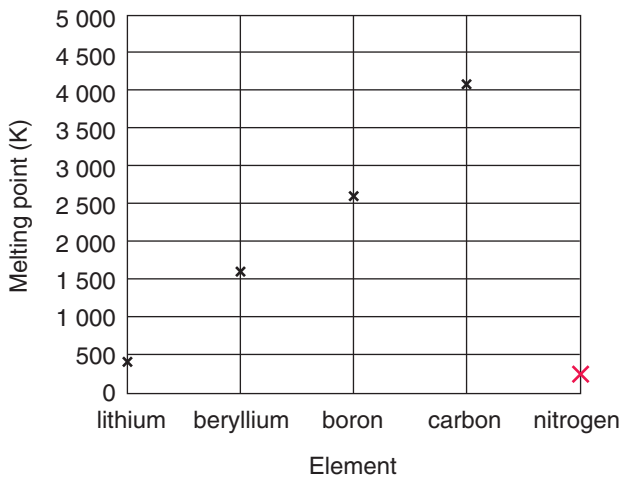
17 State which has the larger atomic radius, a sodium atom or a chlorine atom. Explain your answer.

(AQA Advanced Subsidiary GCE, Unit 1, Jun. 2008, 4(a))

18 Trends in physical properties occur across all periods in the periodic table. This question is about trends in the Period 2 elements from lithium to nitrogen.

a) Identify, from the Period 2 elements lithium to nitrogen, the element that has the largest atomic radius. Lithium

b) Draw a cross on the diagram to show the melting point of nitrogen.



c) Explain, in terms of structure and bonding, why the melting point of carbon is high.

Refer to page T14.

(AQA Advanced Subsidiary GCE, Unit 1, Jan. 2012, 2(a), (d)–(e))

19 State the type of structure shown by crystals of sulphur and phosphorus.

Explain why the melting point of sulphur is higher than the melting point of phosphorus.

(AQA Advanced Subsidiary GCE, Unit 1, Jan. 2011, 5(b))

19 Simple molecular structures
Sulphur exists as S_8 molecules while phosphorus as P_4 molecules.
The size of a S_8 molecule is larger than a P_4 molecule. / There are more electrons in a S_8 molecule than a P_4 molecule.
Hence the van der Waals' forces between sulphur molecules are stronger than those between phosphorus molecules.
So, more heat is needed to separate the sulphur molecules than the phosphorus molecules during melting.

20 The melting points of the elments of Period 3 are given in the table below. Use these values to answer the questions that follow. Refer to page T15.

Element	Melting point (K)
Na	371
Mg	923
Al	933
Si	1 683
P (white)	317
S (monoclinic)	392
Cl	172
Ar	84

a) Explain why the melting point of sodium is very much less than that of magnesium.

b) Explain why the melting point of silicon is very much greater than that of white phosphorus.

c) Explain why the melting point of argon is the lowest of all the elements of Period 3.

d) Explain why magnesium is a good conductor of electricity whereas sulphur is a non-conductor.

(Edexcel Advanced Subsidiary GCE, Unit 1, Jan. 2012, 23)

21 Lithium, beryllium, carbon (graphite) and nitrogen are elements of the second period of the periodic table. Arrange them in increasing order of melting point, and explain the order in terms of structure and bonding.

(Communication mark will be awarded for this question.)

(HKDSE, Paper 1B, 2013, 13)

22 Consider the following oxides:

Na_2O MgO Al_2O_3 SiO_2 P_4O_{10} SO_2 Cl_2O

a) Which of the oxides listed above can conduct electricity in molten state?

b) Explain why SiO_2 has the highest melting point among the covalent oxides listed above.

c) Write a chemical equation for the reaction between $Al_2O_3(s)$ and NaOH(aq).

(HKDSE, Paper 1B, 2012, 16)

23 The Group II element barium (Ba) is silvery white when pure but blackens when exposed to air. The blackening is due to the formation of both barium oxide and barium nitride. Refer to page T15.

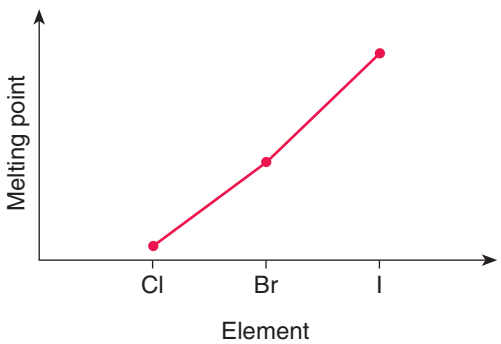
a) Draw an electron diagram of barium nitride.

b) Barium reacts with water to form barium hydroxide solution and hydrogen.

i) Write a chemical equation for the reaction.

ii) State the approximate pH of the barium hydroxide solution.

c) Sketch the trend in melting point of the three Group VII elements, Cl, Br and I. Account for the trend.



d) The Group II elements become more reactive down the group. The Group VII elements become less reactive down the group.

Explain this difference between Group II and Group VII.

24 Both sodium and chlorine are elements in Period 3 of the periodic table.

a) At room temperature and atmospheric pressure, Na_2O is a solid with a very high melting point whereas Cl_2O is a gas. Account for this difference in property between Na_2O and Cl_2O.

b) Na_2O_2 is another oxide of sodium. Write the chemical equation for the reaction of Na_2O with water and that of Na_2O_2 with water.

(HKASLE, Paper 2, 2008, 2(b)(i)–(ii))

25 a) Complete the table below. Refer to page T16.

Element	Na	Mg	Al	Si	P	S	Cl
Formula of common oxide							
Chemical bonding of oxide							
Structure of oxide							

b) Gallium (Ga) belongs to Group III while arsenic (As) belongs to Group V of the periodic table. The table below lists the melting points of oxides of these two elements.

Oxide	**Melting point (°C)**
Ga_2O_3	1 795
As_2O_5	315

i) Suggest the likely bonding and structure of each oxide.

ii) Predict whether or not each oxide will react with

(1) NaOH(aq);

(2) HCl(aq).

Write appropriate equations for all reactions that occur.

26 Account for the following:

$Al_2O_3(s)$ is soluble in both aqueous acids and aqueous alkalis.

(HKASLE, Paper 2, 2010, 5(c))

27 The data in the table below show the melting points of oxides of some Period 3 elements.
Refer to page T16.

	Na_2O	P_4O_{10}	SO_2
T_m (K)	1 548	573	200

a) In terms of structure and bonding, explain why

i) sodium oxide has a high melting point;

ii) sulphur dioxide has a low melting point.

b) Explain why the melting point of P_4O_{10} is higher than the melting point of SO_2.

c) Write the chemical equations for the reactions of Na_2O and P_4O_{10} with water. In each case give the approximate pH of the resulting solution.

d) Write the chemical equation for the acid-base reaction that occurs when Na_2O reacts with P_4O_{10} in the absence of water.

(AQA Advanced GCE, Unit 5, Jan. 2012, 3)

28 Sodium, magnesium, aluminium and silicon are successive elements in Period 3. The table below lists their relative electrical conductivities.

Element	**Relative electrical conductivity**
Sodium	1.00
Magnesium	1.03
Aluminium	1.75
Silicon	0.00

a) Explain why the electrical conductivities of the metals are in the order:

Na < Mg < Al

b) Account for the poor electrical conductivity of silicon.

28 a) Going from Na to Al, the number of delocalized electrons increases. There are more electrons available to drift through the structure when the metal conducts a current.
So, the electrical conductivity increases.
b) In silicon, the four outermost shell electrons in each atom are held in four covalent bonds.
At room temperature, very few electrons gain enough energy to become delocalized.

29 a) Explain why phosphorus pentoxide has a higher melting point than sulphur dioxide.

b) Write the chemical equation for the reaction of P_4O_{10} with water to form phosphoric acid.

c) A waste-water tank was contaminated by P_4O_{10}. The resulting phosphoric acid solution was neutralized using an excess of magnesium oxide. The mixture produced was then disposed of in a lake.

i) Write the chemical equation for the reaction between phosphoric acid and magnesium oxide.

ii) Explain why an excess of magnesium oxide can be used for this neutralization.

iii) Explain why the use of an excess of sodium hydroxide to neutralize the phosphoric acid solution might lead to environmental problems in the lake.

(AQA Advanced GCE, Unit 5, Jun. 2012, 1(c)–(e))

29 a) A P_4O_{10} molecule is larger than a SO_2 molecule. / There are more electrons in a P_4O_{10} molecule than in a SO_2 molecule. Hence the van der Waals' forces between P_4O_{10} molecules are stronger than those between SO_2 molecules.
So, more heat is needed to separate the P_4O_{10} molecules during melting.

b) $P_4O_{10}(s) + 6H_2O(l) \longrightarrow 4H_3PO_4(aq)$

c) i) $3MgO + 2H_3PO_4 \longrightarrow Mg_3(PO_4)_2 + 3H_2O$
or $MgO + 2H_3PO_4 \longrightarrow Mg(H_2PO_4)_2 + H_2O$
or $MgO + H_3PO_4 \longrightarrow MgHPO_4 + H_2O$

ii) MgO is slightly soluble / weakly alkaline.

iii) An excess of sodium hydroxide would make the lake alkaline / kill wildlife.